

UAS Research Program Operator Training Framework

Bridging the gap between Part 107 certification and research-capable UAS operation. For faculty and program directors.

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About This Framework

For faculty principal investigators, aviation program directors, and lab managers structuring operator training for research programs. Covers the capability gap between regulatory certification and research-ready operation, and provides a structured pathway to close it.

The Part 107 Gap

Part 107 Certifies	Research Operations Require
Written knowledge exam (airspace, weather, regulations)	Independent platform operation without support staff
Aeronautical decision-making concepts — no practical test	Field repair and pre-flight troubleshooting capability
No platform-specific training required	Platform-specific configuration and failure analysis
No proficiency evaluation — exam only	Documented proficiency defensible to IRB and institution
No training on autonomous systems or edge computing	Operator-supervised autonomy for AI-enabled research platforms

Sample Training Pathway for Research Assistants

Phase	Requirement	FFA Course	Outcome
Phase 1 — Foundation	FAA Part 107 certification	FFA-201 prep or self-study	Regulatory compliance baseline
Phase 2 — Platform Ops	Platform-specific operations and failure analysis	FFF-401 or FFA-401	Independent operation, field repair
Phase 3 — Specialization	Domain capability (RF, electronics, AI)	FFR-101, FFF-201, or FFA-401	Research-specific platform proficiency
Phase 4 — Documentation	Capability assessment for IRB and program records	Assessment Services	1–5 ratings across 5 domains, IRB-defensible

Course Integration Guide

How FFA courses map to university research program requirements. Academic pricing on all courses.

FFA-201	Autonomy Fundamentals for Operators Foundational autonomy concepts for operators entering AI-enabled research. No prior ML or CS background required. 2 days Academic: \$3,000/student
FFA-401	Operational Edge Computing for UAS Deploy real-time AI on Jetson Orin Nano — one per student, hardware retained. For robotics, aerospace, and CS research labs. 5 days Academic: \$4,000/student Hardware included & retained
FFF-201	Deployed Electronics Repair & Diagnostics Component-level repair, soldering, and failure analysis. Builds lab capability to sustain research platforms without vendor support. 5 days Academic: \$5,500/student
FFF-301	Additive Manufacturing for Sustainment 3D printing and rapid prototyping for platform repair and modification. For any lab that runs platforms in the field. 3 days Academic: \$3,000/student
FFR-101	RF Fundamentals & Spectrum Awareness Baseline RF literacy for research operators. Understand link behavior, interference, and communications resilience without an engineering background. 2 days Academic: \$3,000/student

Sample Flight Test Protocol Checklist

PRE-MISSION (DAY BEFORE)

- ☐ Airspace authorization obtained (LAANC or waiver as required)
- ☐ Weather briefing reviewed — wind, precipitation, visibility within limits
- ☐ Platform inspected — motors, props, battery, frame integrity
- ☐ Emergency procedures briefed to all personnel on site

DAY-OF PRE-FLIGHT

- ☐ Battery voltage and health verified
- ☐ Flight controller calibration confirmed (compass, IMU)
- ☐ RC link and telemetry tested at ground level
- ☐ GPS satellite acquisition verified (minimum 8 satellites)
- ☐ Research payload power-up and data logging confirmed

POST-FLIGHT

- ☐ Flight log downloaded and archived
- ☐ Battery discharged to storage voltage if not flying within 24 hours
- ☐ Any anomalies documented in maintenance log

Ready to Train Your Research Team?

Tell us about your program, platforms, and timeline. Custom training package and proposal returned within 48 hours. Academic pricing on all 10 courses.

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